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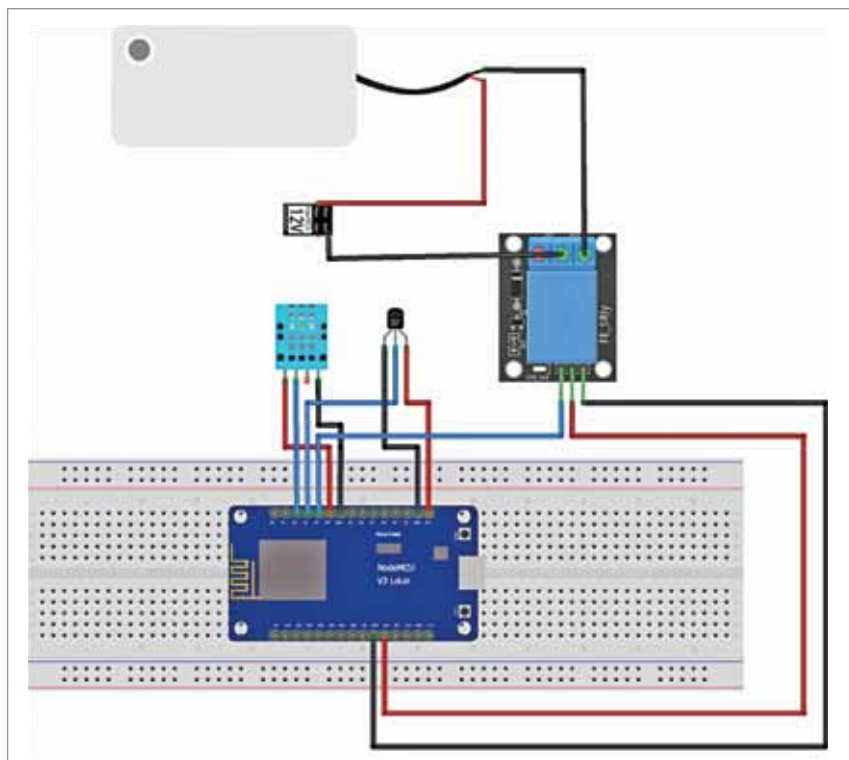


Fig. 7: Wiring diagram of the proposed setup on the breadboard

TABLE 2
BOARD LABELS AND PIN FUNCTIONS OF ESP8266

Connections for DHT11		
+	Voltage in	3V3
Out	Output of sensor	D2
-	Ground	GND
Connections for DS18B20		
V	Voltage in	3V3
D	Output of sensor	D3
G	Ground	GND
Connections for relay		
Vcc	Voltage in	3V3
GND	Ground	GND
IN1	Input for relay	D4

around the ESP8266 module, the temperature and humidity sensor DHT11, the water temperature sensor DS18B20, a two-channel relay module, and a peristaltic pump. Fig. 7 illustrates the wiring connections for the same. Connect the components accordingly, referring to Fig. 6, Fig. 7, and Table 2.

Testing

Power the components and ESP, and wait for some time. The ESP8266 will connect to the network and display the time and sensor readings in the serial port. It will also turn on the water valve and sprinkle water at the scheduled time.

In conclusion, it is possible to distribute the shower water to each and every plant in the hydroponics setup using the 4-way fogger with the modified valve and peristaltic pump. The system operates effectively, providing water showers lasting for tens of seconds within a minute of operation, specifically from 16:00 to 16:01 hours. **EFY**

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